

Homophily in Congress on Social Media

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Introduction

For our analysis, we are concerning ourselves with the [Twitter Interaction Network for US Congress](#) by the SNAP. It represents Twitter activity among members of the 117th U.S. Congress. The dataset contains 475 congressional Twitter accounts connected by 13,289 directed, weighted edges. An edge represents an observed interaction between two members, including retweets, quote tweets, replies, and mentions. Edge weights capture the relative strength of these interactions and are modeled as a normalized measure of influence between accounts. As the dataset itself does not contain any node properties besides the names and usernames of the members, we are enriching it using data from [voteview](#) and [github/unitedstates](#) which cover chamber and party affiliation, age, as well as ideological position scores.

Our focus of analysis will be to assess homophily within the interaction network, based on different characteristics such as party affiliation, chamber and age. It has been shown quite thoroughly that party homophily exists, but we are most interested in whether tertiary factors such as chamber and age moderate this effect, e.g. do members of the senate exhibit more homophily than those in the house, and, do older members exhibit more or less homophily than their younger colleagues?

Theory

Homophily is the principle that contact between similar people occurs at a higher rate than among dissimilar people. It can be further distinguished in two types: *status homophily* as the similarity of persons in their formal, informal or ascribed status and *value homophily* based on shared beliefs, values and attitudes of persons (McPherson, Smith-Lovin, and Cook 2001). We take the party affiliation, the chamber and the age as status homophily and expect that the number of ties between members of congress is shaped by their status, e.g. congress members mostly interact with members of their own party. Age is expected to further shape the number of connections, since people tend to form ties with those of similar life stage (McPherson, Smith-Lovin, and Cook 2001), older legislators may have formed cross-party relationships before the

current era of polarization, but also older members may tend to be less active on twitter. Also a legislators chamber structures the opportunity for interaction, as Senate and House members operate in largely separate institutional environments with different rules, schedules, and legislative responsibilities, making cross-chamber ties less likely to form regardless of party. Value homophily is captured as left-right ideological position and stance on social and racial issues. There is a considerable tendency for adults to associate with those of their own political orientations (McPherson, Smith-Lovin, and Cook 2001). Therefore, we expect to find more interactions between members with similar positioning in the ideological dimensions.

Assortativity is the network-level measure that formally captures homophily. Newman (2003) defines the assortativity coefficient as a Pearson correlation computed over all edges, ranging from -1 to 1 , where 1 indicates that all edges are within-group. In the context of the congressional Twitter network, assortativity provides a useful way to evaluate whether interaction patterns reflect political and demographic divisions. We therefore use assortativity to assess whether members cluster by party affiliation, chamber, age, and ideological position, and to compare the relative strength of these different forms of homophily within the network.

Data Pre-Processing

Exploratory Data Analysis

Overall, the network is moderately sized and active, but not densely connected. Members form many retweet ties in absolute terms, yet these ties represent only a small share of all possible member-to-member connections ($\approx 5\%$). The typical congress member has around 22–24 connections, while a small number of members have far higher connectivity. This suggests a selective and uneven retweet structure, with activity concentrated around more central or more active members rather than spread uniformly across the network. This is to be expected given in politics there are often central figures that may receive more attention, such as presidents or speakers.

```
# Core descriptives: graph size

n_vertices <- vcount(g)
n_edges    <- ecoun(g)
density    <- edge_density(g)

size_summary <- tibble::tibble(
  Measure = c(
    "Number of vertices",
    "Number of edges",
    "Edge density"
  )
)
```

```

),
Value = c(
  n_vertices,
  n_edges,
  density
)
) %>%
dplyr::mutate(Value = round(Value, 3))

cat(
  knitr::kable(
    size_summary,
    format = "pipe",
    col.names = c("Measure", "Value")
  ),
  sep = "\n"
)

```

Measure	Value
Number of vertices	475.000
Number of edges	13289.000
Edge density	0.059

```

# Degree and weighted degree / strength

deg_in  <- degree(g, mode = "in")
deg_out <- degree(g, mode = "out")

str_in  <- strength(g, mode = "in")
str_out <- strength(g, mode = "out")

degree_summary <- tibble::tibble(
  Measure = c(
    "In-degree",
    "Out-degree",
    "Weighted in-strength",
    "Weighted out-strength"
  ),
  Min = c(
    min(deg_in, na.rm = TRUE),

```

```

    min(deg_out, na.rm = TRUE),
    min(str_in, na.rm = TRUE),
    min(str_out, na.rm = TRUE)
  ),
  Q1 = c(
    quantile(deg_in, 0.25, na.rm = TRUE),
    quantile(deg_out, 0.25, na.rm = TRUE),
    quantile(str_in, 0.25, na.rm = TRUE),
    quantile(str_out, 0.25, na.rm = TRUE)
  ),
  Median = c(
    median(deg_in, na.rm = TRUE),
    median(deg_out, na.rm = TRUE),
    median(str_in, na.rm = TRUE),
    median(str_out, na.rm = TRUE)
  ),
  Mean = c(
    mean(deg_in, na.rm = TRUE),
    mean(deg_out, na.rm = TRUE),
    mean(str_in, na.rm = TRUE),
    mean(str_out, na.rm = TRUE)
  ),
  Q3 = c(
    quantile(deg_in, 0.75, na.rm = TRUE),
    quantile(deg_out, 0.75, na.rm = TRUE),
    quantile(str_in, 0.75, na.rm = TRUE),
    quantile(str_out, 0.75, na.rm = TRUE)
  ),
  Max = c(
    max(deg_in, na.rm = TRUE),
    max(deg_out, na.rm = TRUE),
    max(str_in, na.rm = TRUE),
    max(str_out, na.rm = TRUE)
  )
) %>%
dplyr::mutate(
  dplyr::across(where(is.numeric), ~ round(.x, 3))
)

cat(
  knitr::kable(
    degree_summary,

```

```

    format = "pipe"
  ),
  sep = "\n"
)

```

Measure	Min	Q1	Median	Mean	Q3	Max
In-degree	0.000	13.000	22.000	27.977	37.000	127.000
Out-degree	1.000	17.000	24.000	27.977	35.000	210.000
Weighted in-strength	0.000	0.058	0.117	0.163	0.204	1.648
Weighted out-strength	0.002	0.080	0.134	0.163	0.196	0.944

```

# Degree distributions

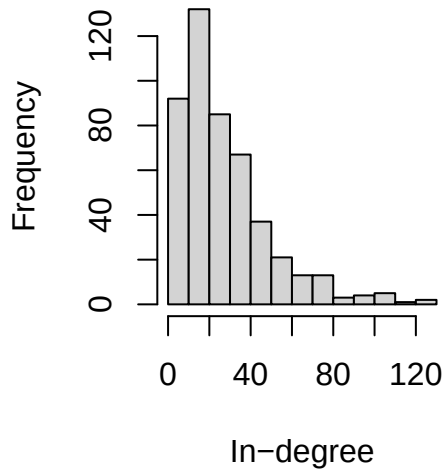
par(mfrow = c(1, 2))

hist(
  deg_in,
  main = "In-degree distribution",
  xlab = "In-degree"
)

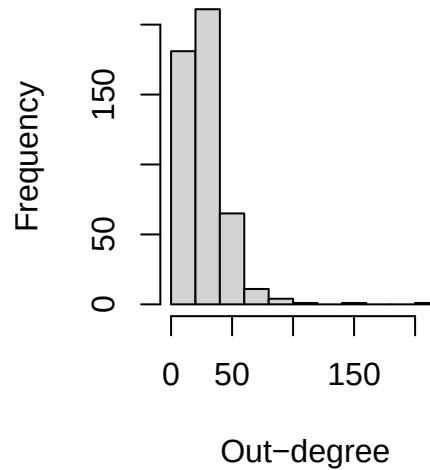
hist(
  deg_out,
  main = "Out-degree distribution",
  xlab = "Out-degree"
)

```

In-degree distribution



Out-degree distribution



```
par(mfrow = c(1, 1))

# Structural network measures

recip <- reciprocity(g)

trans_global <- transitivity(g, type = "global")
trans_avg <- transitivity(g, type = "average")

assort_degree <- assortativity_degree(g, directed = TRUE)

structure_summary <- tibble::tibble(
  Measure = c(
    "Reciprocity",
    "Global transitivity",
    "Average local transitivity",
    "Degree assortativity"
  ),
  Value = c(
    recip,
    trans_global,
    trans_avg,
    assort_degree
  )
) %>%
  dplyr::mutate(Value = round(Value, 3))
```

```
cat(
  knitr::kable(
    structure_summary,
    format = "pipe",
    col.names = c("Measure", "Value")
  ),
  sep = "\n"
)
```

Measure	Value
Reciprocity	0.462
Global transitivity	0.270
Average local transitivity	0.301
Degree assortativity	-0.096

The network shows a moderate level of reciprocity: about 46% of directed ties are reciprocated. This suggests that interactions are not purely one-sided, but often occur between pairs of members who engage with each other, possibly congress members working on the same topics or from similar geographic regions. Transitivity is also moderate, with global and average local clustering around 0.27–0.30, indicating that interaction patterns are somewhat clustered within groups of mutually connected members. This is to be expected given we are not accounting for chamber affiliation and or party affiliation at the moment, so clustering is to be expected. The slightly negative degree assortativity (-0.096) suggests that members with many outgoing interactions are not especially likely to direct those interactions toward members with many incoming interactions. In other words, highly active congress members are not disproportionately interacting with highly visible or frequently targeted congress members.

```
# Homophily by party
el <- igraph::as_data_frame(g, what = "edges")
el$from_party <- V(g)$party[match(el$from, V(g)$name)]
el$to_party <- V(g)$party[match(el$to, V(g)$name)]
mix <- xtabs(weight ~ from_party + to_party, data = el)
mix_prop <- mix / rowSums(mix)
cat("Overall\n", knitr::kable(round(mix_prop, 3), format = "pipe"), sep = "\n")
```

Overall

	Democrat	Independent	Republican

Democrat		0.878	0.002	0.120
Independent		0.770	0.000	0.230
Republican		0.118	0.001	0.881

We can see very strong party homophily: Democratic members mostly interact with Democrats (0.878), and Republican members mostly interact with Republicans (0.881). Cross-party interaction is much lower, around 0.12 in both directions. The “Independent” category is too small to interpret. As we can see below, there does not seem to be much difference in these homophily structures between the house and senate.

```
# Homophily by party by chamber
el$from_chamber <- V(g)$chamber[match(el$from, V(g)$name)]

for (ch in c("House", "Senate")) {
  el_ch <- el[!is.na(el$from_chamber) & el$from_chamber == ch, ]
  mix    <- xtabs(weight ~ from_party + to_party, data = el_ch)
  mix_prop <- mix / rowSums(mix)
  cat(ch, "\n", knitr::kable(round(mix_prop, 3), format = "pipe"), sep = "\n", "\n")
}
```

House

		Democrat	Independent	Republican
:-----		-----:	-----:	-----:
Democrat		0.886	0.001	0.113
Republican		0.103	0.000	0.897

Senate

		Democrat	Independent	Republican
:-----		-----:	-----:	-----:
Democrat		0.855	0.003	0.142
Independent		0.770	0.000	0.230
Republican		0.189	0.004	0.806

Left-right ideology is moderately strong (0.447), meaning members tend to interact with ideologically similar members. Assortativity on the social/racial dimension is weaker (0.119), and age assortativity is essentially negligible (0.038) which means that congress members do not interact more often with members of their age.

```

# assortativity
leftrightass <- assortativity(g, V(g)$nominate_dim1, directed = TRUE) # L/R
socialracialass <- assortativity(g, V(g)$nominate_dim2, directed = TRUE) # social/racial issues
ageass <- assortativity(g, V(g)$age24, directed = TRUE) # age -> basically none

assort_table <- data.frame( dimension = c("Left-right", "Social/racial issues", "Age"), assortativity = c(leftrightass, socialracialass, ageass))

cat(
  knitr::kable(
    assort_table,
    digits = 3,
    col.names = c("Dimension", "Assortativity"),
    format = "pipe"
  ),
  sep = "\n"
)

```

Dimension	Assortativity
Left-right	0.447
Social/racial issues	0.119
Age	0.038

```

# Homophily by age
el$same_party <- !is.na(el$from_party) &
  !is.na(el$to_party) &
  el$from_party == el$to_party

# filter NAs
el_known <- el %>% filter(!is.na(from_party), !is.na(to_party))
# exclude Independents
el_known <- el_known[el_known$from_party != "Independent" & el_known$to_party != "Independent", ]

# per-node homophily score
homophily_score <- el_known %>%
  group_by(from) %>%
  summarise(total_w = sum(weight), ingroup_w = sum(weight[same_party]), h_score = ingroup_w / total_w)

homophily_score$age <- V(g)$age24[match(homophily_score$from, V(g)$name)]
homophily_score$party <- V(g)$party[match(homophily_score$from, V(g)$name)]

```

```
# correlation
cat("Correlation between age and homophily:\n",cor(homophily_score$h_score, homophily_score$
```

```
Correlation between age and homophily:
-0.03447944
```

```
# by party --> democrats weakly positive (+0.04), republicans kind of substantially negative
homophily_score %>%
  group_by(party) %>%
  summarise(`Correlation Age <-> Party Homophily` = cor(h_score, age, use = "complete.obs"))
mutate(`Correlation Age <-> Party Homophily` = round(`Correlation Age <-> Party Homophily`
{ cat(knitr::kable(., format = "pipe"), sep = "\n") }
```

party	Correlation Age <-> Party Homophily
Democrat	0.048
Republican	-0.128

Here, we are looking at whether the party homophily of individuals is higher or lower depending on their age. The correlation is very small and slightly negative (-0.034) overall, suggesting that older and younger members do not differ much in their tendency to interact within party. By party, the relationship is weakly positive for Democrats (0.048) and modestly negative for Republicans (-0.128), indicating that older Republican members are somewhat less party-homophilous in their tweet interactions.

Visualisation

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Conclusion

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References

- McPherson, Miller, Lynn Smith-Lovin, and James M. Cook. 2001. “Birds of a Feather: Homophily in Social Networks.” *Annual Review of Sociology* 27 (1): 415–44. <https://doi.org/10.1146/annurev.soc.27.1.415>.
- Newman, M. E. J. 2003. “Mixing Patterns in Networks.” *Physical Review E* 67 (2): 026126. <https://doi.org/10.1103/PhysRevE.67.026126>.